

Weekly Recap: Shuttle Path Work

Kinematic Work

I built splines from the Excel data and extracted two path sets:

- Small loop points
- Large loop points

These points were exported to CSV and used to drive the shuttle in kinematic mode (`set_pose`). Roll and pitch are fixed at zero; yaw is estimated from the path tangent between consecutive points.

Key Observation

Using only X, Y, Z is not sufficient for fully realistic motion. The yaw approximation makes the rotation look simplified compared to the real shuttle.

Dynamic Attempts

I tested several dynamic-control ideas using the CSV points as force targets, but the behavior stayed unstable and did not converge:

- oscillatory motion and unstable tracking
- unreliable following of the intended path

I decided to pause dynamic work and continue with kinematic mode until the system is functionally complete. Later, I will revisit dynamics in a dedicated branch.

Implementation Notes

- Update period is clamped to 0.02 s for smooth kinematic motion.
- Start poses are aligned so the path passes through the shuttle pose at launch.

Today Meeting (Concise)

We agreed the system should not be modeled as only two fixed loops. Instead, it should be represented as spline-based admissible paths, each tied to a specific switch configuration. The rail topology changes with switch states, so the shuttle is allowed to move only along the physically valid path for the current configuration.

From a control perspective, the switch states define the valid route, and the shuttle can be represented by its progress along the active spline. When switches change, the admissible future path changes accordingly.

If the control program selects a configuration with no valid downstream rail segment, the simulation should not keep the shuttle on a safe trajectory. The constraint should end and the shuttle should fall under gravity, preserving the pedagogical value of the simulation.

Next Steps

Improve path logic to support:

- switch-dependent routing

- path switching based on shuttle position

Implement:

- independent switch control

Prepare architecture for:

- multiple shuttles

- event-based updates

Later:

- revisit dynamic simulation in a separate branch and refine it

Figures

